

# Healthcare & Insurance Opportunity Ideas for PilotFish

Tough industry problems plus common integration patterns not yet covered by the current Sandbox demos.  
Use this as a sales conversation starter, demo backlog, or custom-module / case-study shortlist.

PilotFish wins when traffic is noisy, multi-protocol, heavily transformed, and someone has to own exceptions at 2 a.m. These ideas lean into that — not “call an API once,” but “keep claims, clinical, billing, and payer traffic moving when every trading partner is slightly different.”

## 1. What the Sandbox already demonstrates

Avoid re-building these as net-new ideas unless you deepen them (more SNIP types, real partner endpoints, production ops):

| Demo                        | Domain                  | Core pattern                                   |
|-----------------------------|-------------------------|------------------------------------------------|
| hl7-healthcare-automation   | Payer / hospital HL7    | HL7 ADT batch → validate → SQL + clearinghouse |
| doc-healthcare-hl7-workflow | Gov / behavioral health | Multi-DB poll → HL7 ADT for EHR                |
| medical-lab-hl7-llp         | Lab                     | ORU over LLP → validate → MEDITECH LLP         |
| medical-device-hl7-ehr      | Devices                 | Vitals/CGM ORU LLP → EHR LLP                   |
| fhir-patient-exchange       | Interop / EHR           | FHIR R4 Patient REST create/read               |
| edi-837-snip-sqlserver      | Claims outbound         | SQL claims → 837P + SNIP 1–3                   |
| edi-835-oci-bucket          | Payments / cloud        | 835 → split ST → JSON → OCI Object Storage     |
| sqlserver-pilotfish-demo    | Foundation              | DB poll → XML file export                      |

*Big open surface after those demos: member enrollment (834), eligibility (270/271), claim status (276/277), remittance enrichment, prior auth, quality measures, pharmacy, imaging, worker's comp, ACA / state exchange, and “dirty data” MDM bridges.*

## 2. Tough problems companies struggle to solve

These are the ones that burn budget on one-off scripts, brittle RPA, or “temporary” Point-to-Point connections that become permanent liability. Hard = multi-party, exception-heavy, regulated, or version drift across partners.

### 2.1 Prior authorization that does not die in fax and portal hell

**HARD · PROVIDER + PAYER · FHIR / X12 278**

**Why this hurts:** Payers and providers still lose days to incomplete clinical attachments, portal re-keying, and mismatched code sets (CPT/HCPCS/ICD-10/LOINC). Every payer portal is a snowflake; denial and pended rates stay high.

**PilotFish angle:** Ingest FHIR Questionnaires / Da Vinci PAS bundles (or 278/275), enrich from EHR (FHIR/HL7), attach clinical docs (CDA/PDF/X12 275), validate completeness, route to payer APIs or legacy X12, and return decision status events to the EHR work queue.

**Demo sketch:** Inbound 278 or FHIR PAS → completeness rules → simulated payer decision → 278 response + EHR “auth decided” ORU/SIU notice.

### 2.2 Claims rejection reduction before the clearinghouse

**HARD · RCM · EDI 837 / SNIP+**

**Why this hurts:** Providers discover edits late (clearinghouse or payer). Resubmission cycles destroy revenue-cycle KPIs. Rules differ by payer, plan, and even billing TIN.

**PilotFish angle:** Run SNIP + payer-specific business edits in PilotFish before outbound 837. Kick failures to a work queue with human-readable reasons; only clean claims leave the building. Log every edit outcome to a BI store.

**Demo sketch:** Extend the 837 demo: add a “payer profile” router and synthetic payer edit table that rejects missing referring NPI / invalid POS — show knockout UI.

## 2.3 834 enrollment chaos (employer ↔ broker ↔ payer ↔ TPA)

### HARD · INSURANCE / BENEFITS · EDI 834

**Why this hurts:** Enrollment files arrive late, partially, or with contradictory add/terminate/reinstate logic. Dependent linking, COBRA, and mid-month changes break HR systems and cause denied claims for “not eligible.”

**PilotFish angle:** Normalize multiple inbound 834 dialects (and Excel/CSV from brokers) into a canonical member model; apply term/add reconciliation; emit clean 834/999/TA1 responses plus exception files. Optional: push enrollment deltas to FHIR Coverage / member MDM.

**Demo sketch:** Inbound messy CSV + one 834 → canonicalize → outbound clean 834 + exception report.

## 2.4 Real-time eligibility that is actually reliable (270/271 + FHIR CoverageEligibility)

### HARD · FRONT DESK / PAYER · 270/271 · FHIR

**Why this hurts:** Front desks still get “unable to verify.” Trading partners time out, return partial AAA errors, or contradict portal truth. Caching and fallback logic is ad hoc.

**PilotFish angle:** Front a single eligibility façade: try FHIR CoverageEligibilityRequest, fall back to 270/271, apply timeout/retry/circuit-breaker policy, normalize responses, and cache “known good” windows with audit trails for disputes.

**Demo sketch:** Mock flaky payer + PilotFish retry/fallback showing normalized eligibility JSON to a clinic UI.

## 2.5 Payment integrity / 835 that finance can trust

### HARD · RCM / FINANCE · EDI 835

**Why this hurts:** ERA files show up, but posting fails because CAS codes, PLB adjustments, and split payments do not match open AR. Underpayments hide in noise.

**PilotFish angle:** Beyond “835 → JSON → bucket”: enrich remits with claim lineage, flag underpay thresholds, create denial follow-up work items, and post only confidently matched lines to the PM/EHR with a reconcile report.

**Demo sketch:** Feed sample 835s with underpays; produce matched vs exception buckets + underpay alert CSV.

## 2.6 Quality & HEDIS / Star measure data plumbing

### HARD · PAYER QUALITY · MULTI-FEED

**Why this hurts:** Payers and ACOs chase incomplete clinical evidence across EHRs, labs, and claims. Measure season becomes an ETL panic every year.

**PilotFish angle:** Continuous PilotFish pipelines: claims + ADT + ORU + FHIR Observations into a measure-prep lake/warehouse, with gap-in-care events back to care management (task / SIU / FHIR Task).

**Demo sketch:** Toy Star measure: diabetic A1C gap — claim says diabetic, no recent ORU/Observation → emit care-gap work item.

## 2.7 Provider data management (roster, directory, 274 / FHIR Practitioner)

### HARD · NETWORK OPS · ROSTER / FHIR

**Why this hurts:** Wrong specialty, address, or network status means misdirected referrals and directory inaccuracy (CMS / state scrutiny).

**PilotFish angle:** Ingest rosters from CAQH-like XML, Excel, or payer 274; validate NPI via registry mock; publish FHIR PractitionerRole + directory feed; push deltas to claims adjudication / referral systems.

**Demo sketch:** *Dirty Excel roster → validated FHIR Practitioner bundle + rejection file.*

## 2.8 Interoperability mandates that never stay done (TEFCA / CMS rules / state APIs)

### **HARD · COMPLIANCE · FHIR BULK**

**Why this hurts:** Rulesets and partner onboarding change faster than IT project cycles. Every new endpoint becomes another brittle microservice.

**PilotFish angle:** Use PilotFish as the durable edge: versioned transforms, partner-specific policies, FHIR bulk export/import, and audit logging — so “new regulation” is mostly configuration + route clone, not a rewrite.

**Demo sketch:** *Demonstrate Bulk FHIR NDJSON import with patient matching kickouts.*

### 3. Common high-value demos not covered yet

These are bread-and-butter healthcare/insurance integrations customers expect. They are excellent Sandbox candidates: clear happy path, visible kickouts, and easy executive storytelling.

| Idea                                                     | Difficulty | Why it sells / why build it                                  |
|----------------------------------------------------------|------------|--------------------------------------------------------------|
| EDI 834 enrollment → member DB / FHIR Coverage           | Med        | Every benefits conversation; pairs with 270/271 story.       |
| 270/271 eligibility request/response                     | Med        | Front-desk pain everyone knows; great live demo.             |
| 276/277 claim status inquiry                             | Med        | Closes the loop after 837/835 demos.                         |
| 277CA / 999 / TA1 acknowledgments                        | Low–Med    | Shows productionops maturity, not just “files out.”          |
| ADT + 837 companion: census ↔ claims guardrails          | Med        | Reject claims if no matching admit — concrete \$ story.      |
| CCD/C-CDA → FHIR DocumentReference                       | Med        | Transitions of care / HIE storytelling.                      |
| SIU scheduling ↔ EHR / call center                       | Low–Med    | Visible to operations leaders; easy mock.                    |
| Pharmacy NCPDP or eRx stub                               | Med–Hard   | Opens retail/pharmacy vertical; partner-sensitive.           |
| DICOM worklist / imaging order ORM→ORU                   | Med        | Radiology is still a major PilotFish foothold.               |
| Workers’ comp / property-casualty claim feed (ACORD-ish) | Med        | Insurance beyond health; ACORD/XML often undersold.          |
| Broker portal CSV → canonical → 834 + exceptions         | Low–Med    | Shows “Excel still runs healthcare.”                         |
| SFTP + AS2 + API multi-channel same route family         | Med        | One business process, many intake channels — classic PF win. |

#### 3.1 “Quick win” Sandbox shortlist (suggested order)

- **Eligibility 270/271** — mock payer, clinic UI, AAA error theater, then success.
- **834 enrollment** — CSV + EDI in, clean 834 + exception report out.
- **276/277 claim status** — reuse 837 seed data; status inquiry UI.
- **835 payment integrity** — deepen existing 835 demo with underpay exceptions.
- **C-CDA → FHIR** — one continuity-of-care story for IDN / HIE pitches.
- **Prior auth (278 or FHIR PAS stub)** — harder, but differentiates vs “file mover” tools.

### 4. Vertical slices worth a focused case study

#### 4.1 Behavioral health / Medicaid MCO

##### MEDICAID · MCO

**Why this hurts:** Authorization + encounter data + social determinants from multiple community systems land in incompatible formats. Audits are brutal.

**PilotFish angle:** Extend the DOC-style multi-DB pattern with auth (278), encounters (837I/P), and HL7 ADT into a single Medicaid encounter submission pipeline with missing-data kickouts.

#### 4.2 Dental / vision / specialty carve-outs

##### SPECIALTY PAYER

**Why this hurts:** Carve-out payers speak their own dialects; providers hate portal sprawl.

**PilotFish angle:** Normalize specialty claim input to 837D (or proprietary), apply carve-out edits, return proprietary remits mapped to 835 semantics.

### 4.3 Health system merger / EHR migration bridge

#### IDN · CUTOVER

**Why this hurts:** During Epic/Oracle Health go-lives, temporary interfaces become permanent monsters.

**PilotFish angle:** PilotFish as dual-write / dual-read bridge: legacy HL7 ↔ FHIR ↔ new EHR for a controlled cutover window with replay and compare reports.

### 4.4 Insurance (life / annuity / P&C) new-business and claims

#### LIFE / P&C

**Why this hurts:** ACORD XML, PDF apps, and carrier admin systems still do not line up; STP rates suffer.

**PilotFish angle:** Intake ACORD / eApp JSON/XML → validate → transform to policy admin API + imaging doc store; claim FNOL intake to core claims with acknowledgment SLAs.

## 5. Demo patterns that make PilotFish look uniquely strong

Any middleware can POST JSON. Pitch and demo the failure modes:

- **Exception theater:** show kickouts with readable reasons, not just dead-letter dumps.
- **Partner variance:** two hospitals / two payers, one pipeline, two dialect maps.
- **Protocol bouquet:** SFTP + LLP + REST into the same canonical process.
- **Ack literacy:** 999/TA1/277CA/AA/AE — prove you speak production EDI/HL7.
- **Replay & reprocess:** “fix this claim and send it again” without redeploying code.
- **Observability:** transaction timeline from listener → transform → transport.
- **Custom module honesty:** like OCI Object Storage — show gap → Java module → close gap.

## 6. Conversation prompts for discovery calls

- Which trading partner still requires a human to re-key data every week?
- Where do you lose the most days in prior auth or claim resubmission?
- How many “temporary” Python/PowerShell bridges survived last year’s audit?
- What breaks first when a payer changes an EDI companion guide?
- Can eligibility, claim status, and remittance tell one consistent member/claim story?
- If TEFCA / state exchange / CMS adds another API tomorrow, is that config or a project?

## 7. Suggested next step in this Sandbox

Pick one quick-win demo and one hard differentiator:

| Track           | Candidate                                       | Outcome artifact                                 |
|-----------------|-------------------------------------------------|--------------------------------------------------|
| Quick win       | 270/271 eligibility façade + clinic UI          | Runnable demo + LAN URL + route diagrams         |
| Quick win       | 834 CSV/EDI → member table + exceptions         | Runnable demo + exception PDF sample             |
| Differentiator  | Prior auth PAS/278 completeness + decision loop | Case-study style demo + gaps/custom module notes |
| Deepen existing | 835 underpay / posting exceptions on OCI demo   | Upgrade existing 835 demo rather than new folder |

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